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Maker-centered learning in higher education: a descriptive case study across disciplines at a university makerspace

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Introduction: University makerspaces are increasingly used as arenas for hands-on and collaborative learning, yet empirical studies examining how maker-centered learning unfolds in higher education remain limited, particularly in European contexts. This study investigates how maker-centered learning manifests in courses across disciplines that utilize a university makerspace as a learning environment.

Methods: Using a descriptive multiple-case study design, we examine three courses at Oslo Metropolitan University representing design education, an interdisciplinary engineering project semester, and a first-year engineering mechanics course. Data were collected through qualitative questionnaires from students and course coordinators and analyzed through cross-case thematic analysis.

Results: The analysis identified three recurring sets of learning outcomes across the cases: bridging theory and practice, group work and collaboration, and utilizing the makerspace as a learning arena. Students reported that working with physical artifacts and iterative design processes supported deeper engagement with theoretical concepts, while collaborative project work fostered problem-solving and reflection. At the same time, practical challenges related to materials, tools, and group dynamics formed an important part of the learning process. While many immediate outcomes related to technical skills, these skills were embedded within broader competence development when situated in group-based, problem-oriented, and iterative learning environments.

Discussion: Although based on a limited number of responses from a single institution, the findings provide initial empirical indications of how maker-centered learning may appear in higher education contexts. Interpreted in relation to existing conceptualizations of maker education, the results suggest that university makerspaces can function as environments where creative practice, collaborative inquiry, and hands-on experimentation support competence development across disciplinary boundaries. The study contributes empirical insights into how maker-centered learning may be enacted in university settings and highlights directions for future research on makerspace-based pedagogy in higher education.

KEYWORDS
case study, constructionism, higher education pedagogy, interdisciplinary learning, maker-centered learning, makerspaces, project-based learning

1 Introduction

The Faculty of Technology, Art and Design at Oslo Metropolitan University has established *OsloMet Makerspace*, a creative and transdisciplinary hub for all students and staff at the university. Since its inception in 2018, the makerspace has hosted several courses from diverse disciplines where students are required to apply their theoretical knowledge by creating physical artifacts. However, with one exception (Berg et al., 2020), no courses at the university have been designed with a pedagogical approach that specifically emphasizes maker-centered learning in its curriculum. This highlights the need to better understand how maker-centered learning emerges in courses that use the makerspace as a learning arena and how such practices may inform the pedagogical development of makerspaces in higher education more broadly. In this article, we present a descriptive case study with a multiple-case design to address the following research question: *How does maker-centered learning manifest in courses across disciplines that use a university makerspace as a learning environment?*

As implied in the research question, the theoretical focus of this chapter is the pedagogical approach known as *maker-centered learning*. Rather than being a rigidly defined method, the term maker-centered learning emerged out of Harvard Project Zero's Agency by Design (AbD) initiative (Clapp et al., 2016) as a framework and set of dispositions to support students' development of agency, creativity, and problem-solving skills through hands-on engagement with materials, tools, and systems. In a recent publication, we have defined maker-centered learning as "competence development through creative practice in a technology-oriented and community-driven environment" (Vida et al., 2025). One key term here is *competence*, which in Norway's most recent national curriculum LK20 is defined as "the ability to acquire and apply knowledge and skills in order to master challenges and solve tasks in both familiar and unfamiliar contexts and situations" (Utdanningsdirektoratet, 2020). According to this definition, skills are subcomponents of competence, the latter encompassing understanding and the ability to reflect and think critically. We find this distinction important to point out, because the terms skill and competence are often used interchangeably in literature related to maker-centered learning.

The formative domain of maker-centered learning is the *makerspace*, rooted in the *maker movement* (Dougherty, 2012). Makerspaces are peer-driven workspaces where people can come together to design, build, and experiment using tools and technology. Although commonly associated with equipment such as 3D printers, laser cutters, vinyl cutters, CNC milling machines, and electronics workstations, most makerspaces have a mix of high-tech and traditional equipment on offer. For example, many makerspaces feature machine or hand tools for working with wood, textiles, metals, and more. The term makerspace arose out of the maker movement, which is distinguished by a pervasive do-it-yourself mentality, a belief in learning through making, and an emphasis on knowledge sharing within a flat hierarchy.

The maker movement emerged in the early 2000s as digital fabrication technologies—once prohibitively expensive—became more accessible across industries. Universities and employers simultaneously recognized that engineering graduates lacked hands-on experience due to the decline of traditional shop classes toward the end of the twentieth century (Blikstein, 2013). In response to this need, MIT launched the Fab Lab concept in 2001: a standardized, low-cost laboratory model equipped with digital and traditional fabrication tools that could facilitate interdisciplinary collaboration in both academic and community environments. The movement gained cultural momentum when Dale Dougherty founded Maker Media and launched Make: magazine in 2005, followed by the first Maker Faire—a large fair showcasing a combination of science, technology, and art—in 2006 (Blikstein, 2013). These developments accelerated global expansion of both the maker movement and makerspaces. At universities, makerspaces have become increasingly common as interdisciplinary hubs for creativity and innovation.

Although the maker movement emerged in the entrepreneurial domain, its pedagogical foundations are closely linked to Dewey's pragmatism ("learning by doing"), Piaget's constructivism, and Papert's constructionism (Martinez and Stager, 2019, pp. 11–21). Constructionism, often highlighted as a major influence within maker ideology (Høibo, 2023; Martinez and Stager, 2019), builds on constructivist theory by asserting that students construct new knowledge most effectively when the mental process of understanding is supported through the creation of concrete, personally meaningful artifacts—learning "outside the head" (Martinez and Stager, 2019, p. 36). However, introducing constructionist approaches into educational systems traditionally grounded in *instructionism*—where knowledge is pre-constructed and transmitted by the teacher—remains challenging (Martinez and Stager, 2019, p. 74). Shifting toward constructionism requires repositioning the teacher as a facilitator of learning rather than a transmitter of knowledge (Haakonsen and Skjønneberg, 2020; Raaheim and Nysveen, 2019), and makerspaces offer one fertile arena for such maker-centered learning practices to develop.

Research on makerspaces and maker-centered learning has expanded considerably over the past decade, particularly in North American educational contexts. Systematic reviews of the field show a growing body of empirical work examining the pedagogical potential of maker education, often focusing on creativity, collaboration, and learning through making (e.g., Papavlasopoulou et al., 2017; Schad and Jones, 2020). However, much of this literature is situated in K12 education and in educational systems where the maker movement has had more time to influence pedagogical practices. In contrast, research on maker-centered learning in Nordic educational contexts remains comparatively limited. Høibo (2023) notes that while the maker movement has developed substantial theoretical and pedagogical foundations in countries such as the United States, there is still a lack of research representing a broader diversity of educational contexts. In particular, studies examining how maker-oriented pedagogies develop within Nordic school cultures remain scarce. While this observation also primarily concerns K12 education, the situation appears similar in higher education, where empirical studies of maker-centered learning in university makerspaces remain relatively limited, particularly in European contexts.

To situate our findings within existing research on maker education, we draw on frameworks describing the pedagogical

infrastructures of maker education (Høibo, 2023; Lakkala et al., 2010; Riikonen et al., 2020). These frameworks conceptualize maker-centered learning environments as consisting of several interrelated infrastructures that shape how learning unfolds in practice. Riikonen et al. (2020) identify four key infrastructures —*epistemological*, *social*, *material–technical*, and *scaffolding*—that support learners' engagement in open-ended knowledge creation processes.

The epistemological infrastructure refers to the underlying learning theories informing maker-centered pedagogies. As previously mentioned, maker education is commonly associated with progressive educational traditions that emphasize active knowledge construction through making, experimentation, and reflection, including pragmatism, constructivism, and constructionism. The social infrastructure concerns the collaborative and participatory structures that characterize makerspaces. Maker-centered learning environments typically emphasize peer learning, democratic access to tools and knowledge, and the development of learner agency within collaborative problem-solving processes. The material and technical infrastructure refers to the tools, materials, and technological resources that enable hands-on exploration and prototyping. Through engagement with fabrication technologies and physical materials, learners develop technological understanding, creativity, and practical problem-solving skills. Finally, scaffolding infrastructure refers to the pedagogical supports that enable learners to engage in open-ended exploration. These supports may include project-based learning, interdisciplinary collaboration, mentoring, iterative design processes, and learning environments that treat experimentation and failure as integral components of learning.

In the present study, these four dimensions serve as analytical lenses for interpreting the themes identified in the cross-case analysis. Rather than functioning as a strict coding scheme, the framework provides a conceptual lens for interpreting how different aspects of maker-centered learning appear across the cases examined.

The remainder of this article is organized as follows. Section 2 describes the methodological approach and the design of the multiple-case study. Section 3 presents the findings from the three cases and the cross-case thematic analysis. Section 4 discusses the implications of these findings for understanding maker-centered learning in higher education. Finally, Section 5 concludes the article and outlines directions for future research.

2 Method

This study employs a *descriptive case study* with a *multiple case-design* as its methodological framework. Our approach aligns with Robert K. Yin's definition of case study in which he describes this empirical method as one which

- Investigates a contemporary phenomenon (the "case") in depth and within its real-world context, especially when
- The boundaries between phenomenon and context may not be clearly evident (Yin, 2018, p. 15)

By adopting Yin's definition of the case study, we depart from other empirical research methods such as historical research focusing on non-contemporary phenomena or experimental research separating the investigated phenomenon from its context (Yin, 2018, p. 15). The contemporary phenomenon that we investigate is teaching methods and ways of learning utilizing the university makerspace as a learning arena. The real-world contexts for the investigated subjects are Oslo Metropolitan University and OsloMet Makerspace. We focus on three cases from different disciplines. The first case is from the Department of Art, Design and Drama. The second case is from the Department of Computer Science. Finally, the third case is from the Department of Mechanical, Electrical and Chemical Engineering.

Yin distinguishes between *descriptive*, *explanatory*, and *exploratory* case studies. The purpose of descriptive case studies is to describe a phenomenon in its real-world context (Yin, 2018, p. 287), without necessarily explaining how or why the circumstances occur. The study is therefore framed as a descriptive case study, although it also has an exploratory aim in that it identifies preliminary patterns and themes that may inform future research on maker-centered learning in higher education. The aim is not to establish generalizable claims but to document how maker-centered learning manifests in practice within several courses using the makerspace as a learning arena. Because maker-centered learning is not yet a well-established concept in higher education, especially not in Norway, documenting concrete examples from these cases may help clarify how the approach is enacted in practice and provide a basis for further empirical investigation.

When selecting our cases, we focused on choosing subjects from different disciplines within the Faculty of Technology, Art and Design, where the students were provided with the possibility to work with parts of their projects at OsloMet Makerspace. The courses will be further described in the result section. We gathered data by issuing questionnaires to students who have completed the courses that we have chosen as the focus for our cases, as well as their teachers (from here on referred to as course coordinators). The data from the questionnaires were processed separately for each case and then analyzed in form of a cross-case thematic analysis. The overall design of the study is illustrated in Figure 1.

Although the courses are quite dissimilar in terms of academic fields, the common denominator is that the students have all gone through a period of practical group-based project work in the makerspace. For the students, we devised a questionnaire that was applicable to all cases without needing to customize the questions for the different courses. Our questions focused on the challenges, failures, and successes experienced by the students during their project work in the makerspace, as well as their perceived learning outcomes. A separate questionnaire was issued to the course coordinators, whose roles also included being the lecturer in the core subject or subjects related to the practical project component of the course. This questionnaire focused on the pedagogical aspects of the course and included questions about why the makerspace was chosen as an arena for the practical group work, the balance between theoretical and practical approaches to teaching, and what kind of challenges they perceive that their students faced during the project work.

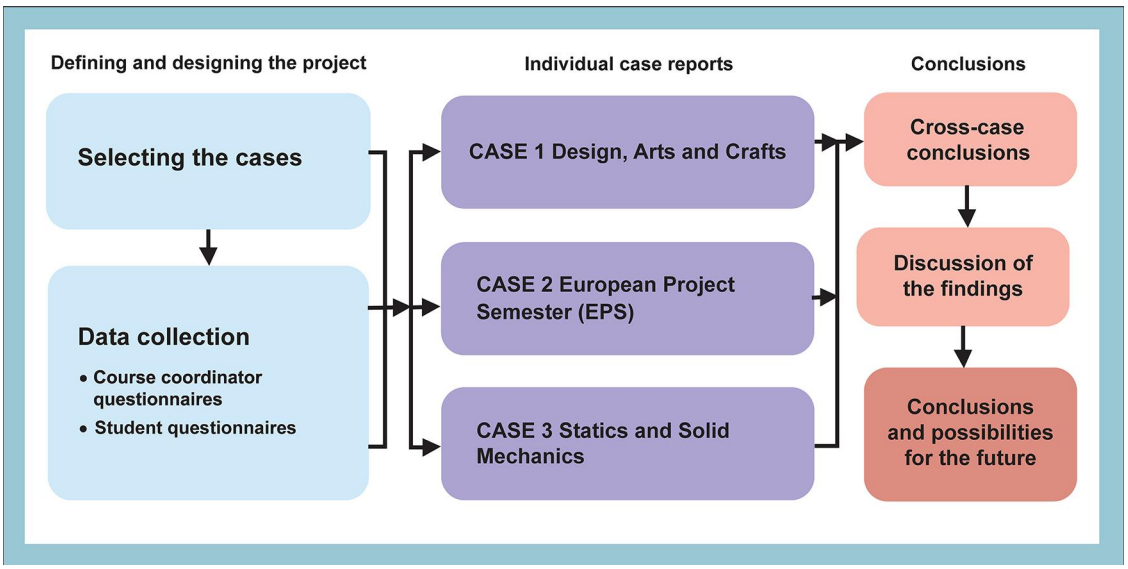


FIGURE 1
Structure for our descriptive case study with a multiple-case design.

Both questionnaires were qualitative, with all questions requiring open-text responses. In our invitations, we estimated that the questionnaire would take 10–15 min. However, this turned out to be a gross underestimation. The average response time was 25 min for the students and 36 min for the course coordinators, albeit with large individual variations. The questions we asked in both questionnaires are listed in Table 1.

We obtained the students' email addresses through the course coordinators, and the questionnaires were embedded as a link in the invitation. As we had already established correspondence with the course coordinators, the target group for our second questionnaire, we invited them to join the survey as part of this correspondence. Both the students and the course coordinators were informed that data would be treated confidentially, and that they would be able to withdraw their participation at any point without any negative consequences. In the case of the course coordinators, full anonymity in the results of the individual cases could not be guaranteed, as it is possible to trace them by looking at the courses' web pages. Prior to the case study, we received approval from Sikt, the Norwegian Agency for Shared Services in Education and Research, which is a requirement for all Norwegian research projects that collect personal information.

3 Case study

3.1 Case report 1: design, arts and craft

3.1.1 Case description

The first case describes the learning experiences of students who have completed entrepreneurship education courses in the program Design, Arts and Crafts at the Department of Art, Design and Drama. This case includes two 10 ECTS Bachelor's courses: *FKH3001 Entrepreneurship and Innovation in Art and Design* and *KD2000 Entrepreneurship in Art and Design*. For brevity, we will refer to the courses only by their codes FKH3001 and KD2000.

We rely on questionnaire responses from a professor who was the course coordinator for both courses, and three students, the first two of which took part in FKH3001 and the third of which took part in KD2000. From here on, the students are referred to as Student 1a, Student 1b, and Student 1c. The course coordinator is referred to as Teacher 1a. Additionally, we support our findings with descriptions of KD2000 by Lena Vida—the second author of this paper—who took over course coordination of KD2000 in 2024.

FKH3001 is a subject affiliated with a former bachelor's program named Specialized Teacher Training in Design, Arts and Crafts, which now has been subsumed by a 5-year master's program called Teacher Education in Practical and Aesthetic Subjects for Years 1–13, Design, Arts and Crafts. This subject has approximately 45 students per course cycle. The teaching methods applied in the course are project- and problem-based learning, and design thinking processes. Additionally, the course utilizes teamwork and the facilitation of real-world projects in collaboration with external partners from the industry, non-profit organizations and public institutions such as galleries, libraries, schools, and public makerspaces. The forms of assessment for the subject include public presentations of the results and a portfolio exam. For the exam, the students need to document the entire design process, from the ideation and research stages via experimentation to prototyping and realization. The report should include self-reflection on the work and a presentation of the acquired learning. Lectures in this course are short, focused, and closely aligned with the curriculum. Due to the range of projects, the supervision that the student groups receive has great importance. Furthermore, group presentations with peer feedback play a crucial role in the course structure. When describing the subject, Teacher 1a

TABLE 1 List of questions asked in both questionnaires.

Questionnaire for students	Questionnaire for course coordinators
Part 1 - Background	Part 1 - Background
What is the name of the course you have taken?	What is the name of the course you coordinated?
Which year/semester?	How many ECTS credits does the course yield?
Which field of study did you come from when entering the course?	How many students on average does your course admit per semester?
Did you have any other academic background prior to this?	Which fields of study do your students typically come from?
	What is your academic or professional background?
	What kind of pedagogical background do you have? (*Academic training and/or practical experience)*
	What kind of teaching methods did you apply when developing the course?
Part 2 - About the course	Part 2 - About the course
Could you describe your student project? *(What did you make? What was your role? What did you learn?)*	In your course, how much weight do you place on the importance of lectures vs. supervision?
What kind of challenges did you encounter during the part of the project that was done in Makerspace? *(Which mistakes were made? What were the next steps? What were the consequences? How did you solve the problems that arose? Did you solve them as a group or individually? What did you learn in the process?)*	How do you manage the balance between theoretical learning and hands-on making in your course?
Could you describe how designing a physical artifact in Makerspace affected your theoretical competence?	Why was it important to include hands-on making as part of your course?
Did you experience issues related to people coming from different academic fields?	Why did you choose Makerspace as an arena for the course?
Did you experience issues related to people coming from different cultures?	What are the reasons for opting for group work as opposed to individual project development for the course?
Could you briefly describe the level of success for the project as a whole? *(Please disregard the final grade. We are more interested in your personal view, i.e., how happy are you with the result?)*	What kind of challenges do students coming from different knowledge domains typically encounter?
Could you also describe things that did not go well, but that you learned from?	Given your experience with the course, what kind of changes do you deem necessary to increase the learning outcomes for the students?
What are the reasons behind your view of the project as successful/not successful?	Do you have any additional comments you wish to add before ending the questionnaire?
Do you have any additional comments you wish to add before ending the questionnaire?	

states that the incorporated teaching methods correspond strongly with what the teacher calls for the "makerspace concept".

Student 1a and Student 1b both participated in FKH3001. As part of their course project and exam, Student 1a and their group created a teaching resource for high school students, where use of the makerspace as a learning arena was highlighted as a method. The technique they chose to focus on in the project was screen printing. Student 1b was part of the same course a year earlier. During their course period, Student 1b served as the CEO of their student-run startup formed with four other classmates. As part of their course and exam project, the group designed a multifunctional tote bag.

3.1.2 Results

Considering the learning outcomes in the subject, both Teacher 1a, Student 1a and Student 1b commends the good balance between theoretical and practical learning in the course, something which is described by them as essential in practicing design processes, such as design thinking.

Teacher 1a states that in the course

> [s]tudents are expected to employ their already acquired practical competences and connect the work they do with relevant theory. They are also expected to experiment with practical processes they do not have experience from and build competences in new areas, for instance in prototyping and testing.

One of the acquired practical competences that Student 1a describes is understanding and appreciation of materiality. Student 1a states that

> [t]he materiality of holding something in your hands, I think, also carries a lot of valuable information for such a process. In a way, I think [it] is underestimated in the same way the

> knowledge that goes into crafts is underestimated. I my field, the physical and theoretical are never completely separate.

In FKH3001, theoretical and practical work goes not only hand in hand, but as part of iterative design processes. Teacher 1a describes design as a practical profession, and elaborates:

> [To] understand the complexity of a design project it is necessary to practice, experience, fail with style, learn, iterate and move on. There is a wide gap between a dream or idea and the realization of a knowledge based, problem solving solution built on research and understanding of the user as well as the realization possibilities, which could be as diverse as a physical product, a work of art, a workshop or a teaching session.

Teacher 1a, Student 1a and Student 1b observe that the university makerspace has been an important arena for practicing design processes. Teacher 1a describes the choice of using the makerspace as one of the learning arenas for the course:

> Makerspace is the perfect arena for introducing students to solving practical problems by working with different techniques using […] different materials. They can start at a low entrance level and develop at their own pace. The experience building my students have achieved in the makerspace has given them the confidence to dare to experiment and demonstrated to them that they are able to solve a wide range of challenges by using the typical makerspace facilities. Makerspace is a training ground for understanding how products are developed and built.

Students 1a and 1b both recognize the significance of using the makerspace for iterative design processes with practical work. Student 1a describes the opportunity to carry out the entire design process from idea to realization as "empowering". Student 1a recounts using the makerspace as part of their course group project very actively, emphasizing that the student themself also gained knowledge about the makerspace during their project work and acquired skills for laser cutting and soldering. Student 1b similarly recognizes the creation of a physical artifact as a tool for better understanding the theoretical framework, particularly design thinking. Student 1b also noted that the final product was more professional thanks to the opportunity to learn about and use the machines.

Another learning outcome associated with the course is working and executing projects in groups. The way Teacher 1a includes group work in his course is by organizing the students in groups of four or five. The students are urged to map the competences of one another as well as their group as a unit. During the course, the groups would explore new territories, and encounter challenges which they can face more effectively as groups. A key element in the group work is discussing the challenges the groups would encounter. Teacher 1a states that "the team learns more from discussing how to approach a challenge, analyzing why a first prototype does not yet work, reflect and find a solution—supported by the theory which is relevant in their specific project". In addition to internal discussions within the groups, discussion across the groups would also be encouraged. Teacher 1a posits:

> When I have ten groups working on different challenges with different external partners, the sharing between the groups is also important. By facilitating information sharing within the class, all groups get insight into what the others do and build a wider understanding than the group alone can achieve. They can use each other as information source or ask for critical feedback.

Both Student 1a and Student 1b similarly recognize working in groups as an important learning outcome of the course.

Working with design processes and working as part of a group are both associated with challenges by the two students. They, however, also describe how they navigated these challenges and transformed them into meaningful learning outcomes. One of the challenges Student 1a's group faced during their project period was that they, as a group "coming from the arts", felt out of place at the makerspace, which they felt belonged more to people with a background in engineering. They managed to overcome this as a group by spending more time there. Regarding the design process itself, Student 1a describes trials and errors as a natural part of the prototype building process: "If something went well, then it was good, but if it didn't, we tried to figure out why and then used that knowledge in the next prototype. Win-win". When it comes to possible challenges during the project work, Student 1b remembers using a lot of time at the makerspace together with another group member, in order to learn and understand the embroidery machine. The informant emphasizes that they solved all problems that would arise *together*.

Working in groups has also been described as a learning outcome associated with challenges both by Teacher 1a and Student 1b. According to Teacher 1a, one of the challenges associated with group work is not recognizing the value of the competence of others:

> In more homogeneous groups with similar backgrounds, recognizing the values of the group members' competences is normally quite easy. In groups with a wider competence span, for example aesthetic focused product design meeting mechanical engineering, there [are] unfortunately occurrences where some students have problems with acknowledging the value of the competence of the others.

Student 1b states that most of the challenges their group encountered stemmed from working in groups. Student 1b adds that during the course, the group members gained a better understanding of each other's strengths and weaknesses, and the final outcome was positive. Even though Student 1a does not highlight working in groups as a challenge, they state that their group has found a new appreciation for group work and being part of a team.

Overall, Student 1a considers their project to be a success. On an individual level, the student highlights becoming better at design processes, screen printing, and using makerspace tools. On a collective level, the student attributes their group's success

to the screen-printed product they created, which worked well and has been in use ever since. Another factor of success that the student pinpoints is the benefit of group work and group dynamics: "I think our group dynamics was really important, as everyone was motivated and contributed in a way in which we really benefited from collaborating". Correspondingly, Student 1b considers their project to be quite successful and attributes their success to good mentoring, access to advanced and relevant technology, and the networking possibilities in the course.

KD2000—the other course we examined in this case—is a bachelor's level course for second-year students enrolled in the program Art and Design. Originally designed by Teacher 1a, it was recently taken over and partially redesigned by Lena Vida, the second author. KD2000 has approximately 60–70 students in each course cycle. The teaching methods, ways of learning, and forms of assessments Vida applies in this course correspond to the ones applied in FKH3001.

Student 1c participated in KD2000 in 2025, and led a student group that were creating a modular solution system for a fashion library with a focus on maximizing productivity and event opportunities. They made furniture models by using the laser cutter to show their client how the modularity of their product would work. Student 1c highlights that designing a physical artifact was rewarding and aided their visualization of the product. Another learning outcome in the project highlighted by the student is learning a lot about networking.

A challenge that Student 1c met during the project work with the laser cutter was determining the right size for the seating in the modules, which first led to producing more material waste. The solution was then found with math, individually by the student, who states that "I learned to measure twice and cut once". Another challenge arose as part of managing a team. The student notes: "I struggled with managing personalities within the group. As a few members did not quite respect each other, and they would not head to my leadership in the beginning". The group worked through these challenges. Our informant further states: "I learned that my skills lie more in labor rather than management, at least for the time being".

Overall, Student 1c considers the project a successful one that provided them with new contacts and possible job opportunities for the future. "The result was good, the client liked our vision, and we had fun for most of the process", the student adds as a final remark.

3.2 Case report 2: European project semester

3.2.1 Case description

Whereas the first case describes a field of study relatively far removed from what one would typically regard as technological, the second case examines European Project Semester (EPS)—a program designed to train "mainly third year engineering and technology students to carry out project work in international teams" (OsloMet, 2024). However, EPS is multidisciplinary at its core and accepts students with such varying backgrounds as industrial design engineering, architecture, international marketing, and business administration. Running over the course of one semester, the project portion of EPS is a 20 ECTS-yielding course where students are divided into multidisciplinary groups. They are tasked with working as teams to design solutions to complex problems based on real societal challenges. The problems are defined by the groups' supervisors.

In this case study, we rely on questionnaire responses provided by three students as well as the course coordinator. From here on, the EPS students will be referred to as Student 2a, Student 2b, and Student 2c. The EPS course coordinator will be referred to as Teacher 2.

The EPS course draws on the methodologies of student active learning and project-based learning. At the end of the course, the groups deliver a project report and subsequently have group presentations, including an oral defense. They get a group grade for the project. Additionally, they receive individual grades for two semester-long supporting courses in Project Planning and Management and English and Academic Writing. Combined, the supporting courses yield 10 ECTS credits, making the entire EPS program worth 30 ECTS. The stated learning outcomes of EPS are the ability to collaboratively plan, execute, and document interdisciplinary projects using project management methodologies, apply scientific and disciplinary knowledge to solve complex problems, and communicate and reflect critically within diverse teams.

3.2.2 Results

The students are not required to work in the makerspace, but many choose to do so. According to Teacher 2, the introduction of the makerspace resulted in a shift in the teaching staff's expectations regarding the artefacts students were expected to produce, "whether these were simple proof-of-concept or more advanced MVPs (Minimum Viable Products). The makerspace provides not only the technical infrastructure, but also the always-present human resources that offer training and soliciting. Student 2c describes situations where they were stuck and on the verge of giving up but finally overcoming problems with the help of student mentors working in the makerspace.

Student 2a and Student 2b worked together in a group, comprising six students and an equal number of nationalities, that decided to build a model of a cargo ship to demonstrate a wing assisted propulsion modular system (Figure 2). Over the course of several months, they designed the ship using 3D modeling software, 3D printed the parts, sewed the sails, assembled the ship, painted it, and integrated electronics for the moving parts. The individuals in the group who were mechanical engineering students handled the more technical aspects of the building process. One physics student, who acknowledged no experience with modeling, focused on exterior design and the sails. The result was a detailed ship model, approximately one meter in length, and complete with a 3D printed captain standing on the bridge.

Student 2c was part of a group of five students with three different nationalities, who made wearable devices for the legs to measure straight leg length discrepancies between the right and left limbs. Such devices could be used for clinical diagnostics, rehabilitation monitoring, gait analysis, or injury prevention. Because of their background in product design, Student 2c assumed responsibility for designing the casing for electronics,

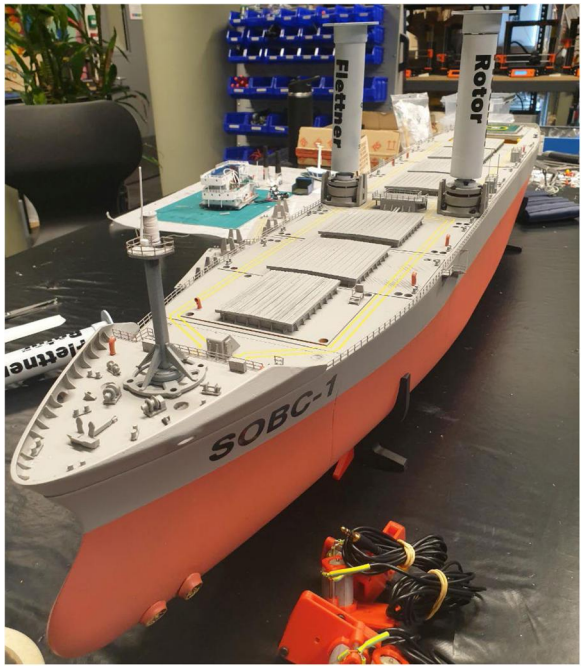


FIGURE 2
Ship building at the makerspace. Photo: Notto J. W. Thelle.

spending several months researching different materials and 3D printing techniques. In the process, the student also had to learn how to solder to optimize the size of the encasing.

Overcoming challenges as a group is one of the most important aspects of the EPS learning experience. The multidisciplinary approach is a fundamental part of addressing these challenges holistically. As Teacher 2 explains it:

> Knowledge in the professional [fields] comes from many different domains; physics, electronics, programming, user-experience. These only manifest as a body of knowledge when they are applied to a problem. Solving the problem allows students to apply their knowledge in a way that [exposes to themselves as much as to us] what they are capable of [and] what they truly understand of their professional know-how.

Incidentally, none of the EPS students claimed any difficulties with working with people from different academic disciplines. On the contrary, they all highlighted the pleasure of discovering how the group members took responsibility for tasks within their own field of expertise. Cultural diversity within the groups was likewise not viewed as an issue, although Student 2c did express some difficulty with understanding English. The reported challenges were mostly of a technical nature. One problem addressed by all students was the experience of multiple failed 3D prints. Student 2b and Student 2c also described the frustration of materials running out and having to wait for new deliveries. Student 2a emphasized the iterative process of discovering design errors when going from virtual models to physical artefacts and having to redesign and try again. According to Student 2b, their ship also broke twice when moving it between locations, leading the group to adopt safety measures for transportation.

The fact that the students did not mention any challenges attributed to their diverse disciplinary backgrounds does not necessarily imply that the challenges were not there, only perhaps that these groups were particularly adept at navigating them through communication. Upon reflection of what kind of challenges students coming from different knowledge domains typically encounter, Teacher 2 explains:

> They have been taught one method as optional for a particular set of problems. First, they attempt to reshape a problem, so it fits their preferred method. Only through discussions (and listening and understanding each other) do they assess different methods and collaborate to decide the best one.

All three students consider their group's results to be successes. Student 2a lauded their group's combined effort as above what both they and their supervisor expected. Student 2c was similarly pleased with their group's final results, emphasizing good teamwork and the opportunity to focus on their own strengths. Student 2a provided a more detailed explanation for the reasons behind their view of the project as a success:

First and foremost, we had a list of objectives to fill. Objectives with very specific goals, which were all completed successfully. However, part of our project would be "subjective", as in we had to build a boat that was good looking, focused on aesthetics. One could consider measuring the success of that hrough the number of people that stop to take a look at the ship and ask about it.

On an ending note, all three of the students also consider the social dimension of working together in the makerspace as an important aspect of their positive experiences. Student 2b emphasizes the fun they had during the project and having made new friends, and Student 2c mentions the "nice workers and the good atmosphere". Along similar lines, Student 2a articulates appreciation for having had the opportunity to discover other cultures as part of the project semester. These comments bring to light the importance of the community aspect of maker-centered learning.

3.3 Case report 3: statics and solid mechanics

3.3.1 Case description

The final case concerns a first-year statics and solid mechanics course at the Department of Mechanical, Electrical and Chemical Engineering. We refer to the course by its acronym MAPE. The data supporting this case study is a combination of the questionnaire responses from four students and one course coordinator in our own survey and secondary literature by Keprate et al. (2023), who have examined the course earlier. The students in our study will be referred to as Student 3a, Student 3b, Student 3c, and Student 3d. The course coordinator is referred to as Teacher 3. Results from Keprate et al. (2023) are included to reinforce the empirical basis of our findings. We explicitly indicate when referencing these secondary findings.

As Teacher 3 explains it, MAPE is by and large a theoretical course, but lectures are complimented with the use of props and voluntary lab assignments for practical understanding. Additionally, the makerspace is used as part of a mandatory model building project, where students are divided into groups of five with the task of designing a crane "capable of lifting a 5 kg weight a minimum of 3 cm, utilizing a 0.5 kg counterweight, while prioritizing lightweight construction" (Keprate et al., 2023). According to Teacher 3, the practical component of MAPE is founded on project-based learning and emphasizes student engagement. A large cohort of students occupies the makerspace for several weeks during the crane-building project. The groups often work at the makerspace at the same time, and there is some inter-group interaction. However, since the project results have an influence on the final grades, the groups are also arguably competitors.

3.3.2 Results

The students are given access to the tools in the makerspace, as well as a limited selection of materials. In most cases, the cranes are largely composed of plywood. The mechanical moving parts are usually 3D-printed or alternatively laser cut from acrylic plates. Strings or fishing lines are used for the hoisting system. Apart from the criteria mentioned above, the task is completely open-ended, and the design is up to the various student groups, resulting in an abundance of cranes in all shapes and sizes. For some groups, visual appeal may also influence design choices. The project component of MAPE enhances the learning outcome by providing the students with first-hand practical experience of the relationship between loads and the strength properties of structures. When something breaks, they learn, and they start again. Teacher 3 posits that the practical element of the course is introduced to make the students fully appreciate the application of the theory they are learning, and to increase engagement. Student 3d seems to corroborate this:

The process of iterative design makes learning the theory a lot more fun. Ultimately giving me more motivation to learn theory that in itself can be somewhat boring.

In terms of challenges, the common denominator among the students who participated in the questionnaire was discovering the importance of the choice of materials or how to configure the equipment for the correct kinds of materials. For instance, Student 3a mentioned that the 3D printer struggled to print out pulleys accurately. When their group finally asked a student mentor for help, they were almost out of time. In hindsight, the student reflected that they could have saved themselves from much trouble by asking for help sooner. Similarly, Student 3d considered that their group's first mistake was the material choice in their first iteration of the crane, which did not have the expected structural integrity. The problem was solved by swapping the failing parts with something sturdier. The student summarizes:

> I learned a lot about iterative design and prototyping. I also learned a lot of the possibilities and limitations of the equipment that [the] makerspace has in their inventory.

These findings are in line with Keprate et al. (2023), who found that the students who failed to produce functioning cranes had not considered the limitations of the materials available to them, and ultimately ran out of time. Those students who had good time management and were able to iterate their design, on the other hand, were all successful in producing cranes that could lift the required weight.

Another challenge the students faced during the practical part of the MAPE project was coming to terms with working in groups that were randomly assigned. For some groups, this led to having to collaborate with people that are not interested in putting in the same amount of work as others. Incidentally, the group challenge was stated as the most challenging part of the project in the paper by Keprate et al. (2023), with many students complaining about not being able to choose their own groups. Although frustrating, Student 3d commented that the positive side of the randomly assigned groups was that the people who were interested got more freedom to do what they wanted, and that the experience of collaborating with different kinds of people with varying levels of motivation was a learning experience. Student 3b expresses a similar sentiment, claiming that their success in the project was due to good communication skills among "the three of us [who] were active in the group".

All the four students who answered the questionnaire considered their projects successful and remarked that designing the cranes increased their theoretical competence. Student 3a claimed that completing the project "effectively showed us some of the theoretical principles being used in practice, making it easier to understand and use further in our studies". Similarly, Student 3b stated that the design process "made it all more understandable", while Student 3d wrote:

> Being able to use a place like [the] makerspace to solve practical problems in our courses really elevated my motivation. For me, it is easier to understand certain topics when presented as a practical, hands-on problem to solve.

The findings of Keprate et al. (2023) once more aligned with our own—most of the students emphasized increased motivation and claimed that the MAPE project helped them to apply theoretical knowledge to practical problems.

3.4 Cross-case thematic analysis

In the individual cases, the course coordinators have described the teaching methods and the forms of learning associated with their courses. These individual accounts show considerable similarities, with project-based learning, problem-based learning and group work as central to each of the presented subjects. Prototyping, learning through iterative design processes, and the possibility for the students to use the university makerspace as a learning arena for their projects has been also utilized in all the presented subjects. Regarding the learning outcomes in the presented cases, there are notable similarities and certain individual traits that will be presented in this section of the paper. We categorize the commonalities from the three cases into three sets of learning outcomes, namely *bridging theory and practice*, *group work and collaboration*, and *utilizing the makerspace as a learning arena* (Figure 3).

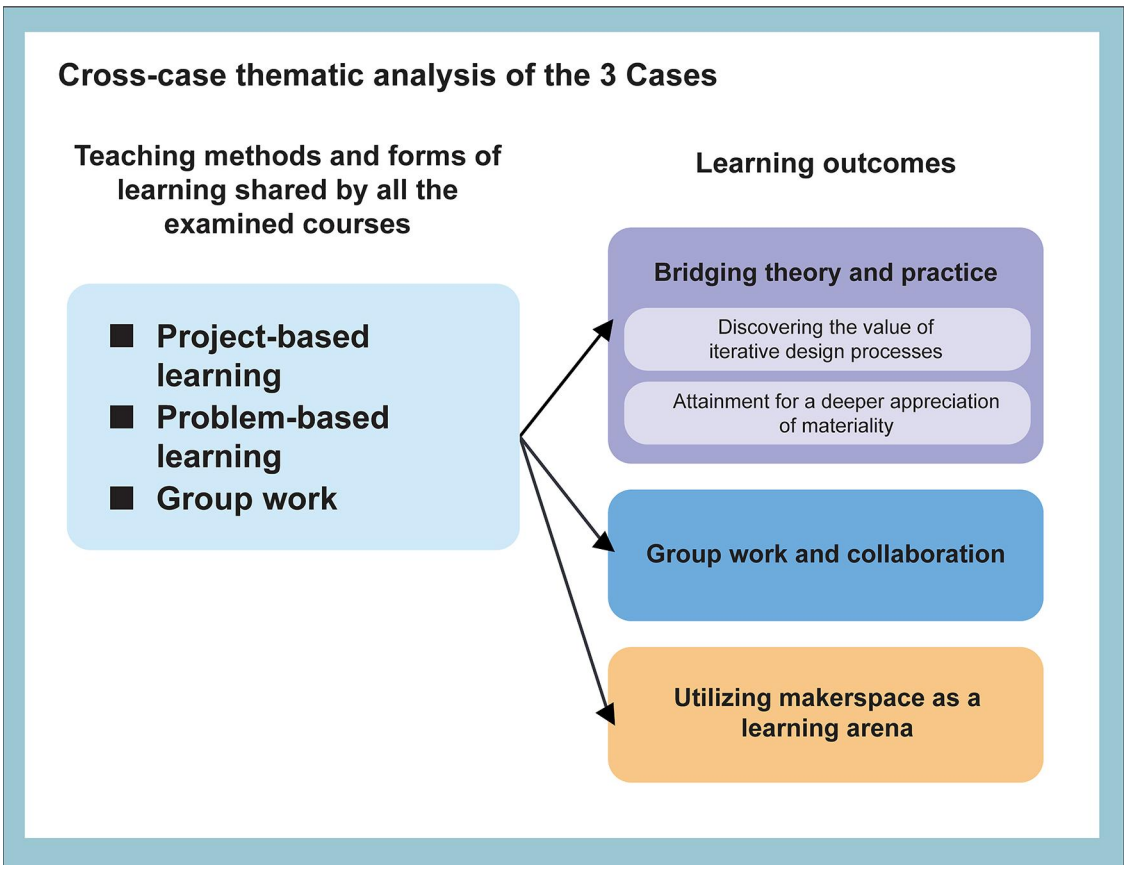


FIGURE 3
Visual representation of the cross-case thematic analysis.

One specific learning outcome falling into the bridging theory and practice set is discovering the value of an iterative design process in realizing ideas, with the physical artifact acting as the articulator of an unformulated idea. It is the enactment of this process that Student 1a describes as empowering. Student 2a similarly emphasizes the value of learning about the iterative process and being able to identify design flaws during the transition from virtual models to physical artifacts. Correspondingly, Student 3d expressed that engaging in hands-on learning and iterative design processes at the makerspace made working with their project more enjoyable, which increased their motivation and made it more comprehensible for the student to learn about the theoretical part of the subject. All students from Case 3 emphasized that being able to work with their projects at the makerspace contributed to their theoretical understanding. Student 3d further elaborated on how doing project work at the makerspace could facilitate a deeper understanding of the theoretical components in their subject.

Another learning outcome in the bridging theory and practice set is the attainment of a deeper appreciation of materiality. For example, Student 1a highlights the knowledge one can gain through "holding something in your hands". Similarly, the students in Case 3 report that they have gained a better grasp of materials while working with their project at the makerspace, particularly by testing the limits of the materials and restarting the process when they break. These insights illustrate how the students link materiality to deeper conceptual understanding, which we interpret as competence rather than mere skill.

Group work and collaboration—the second set of learning outcomes identified across the cases—was initially described as challenges in Case 1 and Case 3. However, these challenges were subsequently worked through and transformed into valuable learning outcomes. Student 1b upholds that group members learning about and understanding each other's strengths and weaknesses was key to their success as a team. Student 1c states that some of the challenges relating to group work leading the group could be attributed to the fact that some of the members did not respect each other or were willing to listen to the person functioning as the leader of the group. This made managing the group challenging at the beginning of their project period. Student 1c explains that, while the team worked through the challenges together, they personally gained a clearer understanding of which roles are better suited to their individual strengths. In Case 2, on the other hand, group work was not described as a problematic. On the contrary, all three students in the case highlighted good collaboration between the group members from the outset as key to their success.

The third set of learning outcomes we have identified—utilizing the makerspace as a learning arena—grew out from initial frustrations with practicalities and logistics. Such struggles included determining the right sizes for parts being cut out on

the laser cutter, having to work through multiple failed 3D prints, configuring 3D-printers correctly, or choosing the right materials. Some of the students discovered that part of working in the makerspace entailed running out of materials and waiting for new deliveries to arrive before being able to continue their projects. These challenges were overcome by spending enough time in the makerspace and getting to know its tools, the materials and the routines. As a case in point, Student 1a described the challenge of their group feeling out of place at the makerspace as arts students, which they overcame by spending more time there.

Although many of the challenges described by students eventually became sources of learning (e.g., mastering the laser cutter after repeated failure, resolving conflicts in group work), they also point to structural issues in the makerspace that may limit the benefits of maker-centered learning. For example, some arts students reported initially feeling out of place in what they perceived as an engineering-dominated environment. This suggests that makerspaces can reproduce existing disciplinary hierarchies as much as they can challenge them, particularly for students who do not already identify with technology-oriented cultures. Similarly, the reports of uneven group participation, difficulties with leadership, and competition between groups in the crane project highlight that makerspace activities can amplify familiar issues in group work and assessment, rather than automatically fostering collaboration. While individual students framed these as important learning experiences, they also raise questions about how responsibility and labor are distributed within student teams.

4 Discussion

The findings from our cross-case analysis provide insights into the learning outcomes that students across disciplines such as arts, crafts, design, and engineering develop when engaging in group-based, problem- and project-oriented work in the makerspace. In this case study, we set out to examine how maker-centered learning is reflected in both the declared learning outcomes of the courses and in the experiences reported by the students. Several of the outcomes identified in the analysis are directly related to using the makerspace as a learning environment. These include developing proficiency in operating machines and associated design software, engaging in iterative design processes, and gaining confidence in navigating the makerspace environment.

In strict terms, we can categorize these as primary learning outcomes falling into the narrow definition of *skills* as opposed to *competence*, with the latter demarcated in the introduction as the application of knowledge and skills in complex, open-ended contexts using reflection and critical thinking. Circling back to the definition of maker-centered learning as "competence development through creative practice in a technology-oriented and community-driven environment" (Vida et al., 2025), we need to identify secondary learning outcomes that may be classified as competences to claim that maker-centered learning is clear in the courses represented in the presented cases. To make the point explicit, becoming a proficient user of a makerspace does not amount to competence building *per se*. Nonetheless, it is a crucial step along the way. We contend that the methods of problem-based or project-based learning, coupled with an emphasis on group work using iterative design processes in the makerspace, together constitute maker-centered learning. One student described the journey from idea to physical artifact as empowering, illustrating the transformative potential of maker-centered learning. By the same token, reflections on the discovery of how working with physical artifacts made students understand theory, as well as on the understanding that group dynamics brings out knowledge that may be hard to attain individually, highlight the pedagogical value of integrating hands-on practice with collaborative inquiry.

Another part of the definition of maker-centered learning we would like to address is the element of creative practice. Outside of the arts- or design-oriented departments, creative practice is generally not regarded as essential. However, our findings suggest that exposing engineering students to project work with a creative component can contribute to deeper theoretical understanding. This aligns with the constructionist stance that knowledge construction can be enhanced when happening "outside the head" in a creative and meaningful process. Thus, maker-centered learning has an inherent transdisciplinary potential, with creative engagement in physical artifacts serving as its catalyst. While our findings indicate that some of the arts students expressed initially feeling out of place in a technology-oriented environment, they became more comfortable as they learned how to use the equipment. Contemporary culture and the way the current education system works tends to reinforce a sense of detachment between creativity and technology, with the latter being strongly associated with scientific-technical disciplines such as computer science and engineering. We posit that this dichotomy is both outdated and counterproductive. Maker-centered learning may help bridge this divide by bringing creative and technical practices into closer dialogue, offering a promising arena for educating students to address complex societal challenges.

This study is based on a relatively small number of questionnaire responses from students and course coordinators at a single institution. The participants were self-selected and may be more positively disposed towards the makerspace than non-respondents. Our data is also retrospective and based on self-report, which limits our ability to assess how makerspace activities influence learning over time or how they compare to other forms of teaching. Accordingly, the findings should be interpreted as initial empirical indications of how maker-centered learning may manifest in higher education contexts, rather than definitive disciplinary characteristics. The positive accounts in our material should therefore be interpreted as examples of what maker-centered learning can look like under favorable conditions, rather than evidence that such outcomes are typical or guaranteed.

Although the themes identified in this study were derived inductively through cross-case thematic analysis, they can also be interpreted in light of existing conceptualizations of maker-centered learning environments. In particular, Riikonen et al.'s (2020) framework of pedagogical infrastructures, introduced earlier in this article, provides a useful point of reference for reflecting on how different aspects of maker-centered learning appear in the cases examined. As noted in the methodological section, this framework is used here as an interpretive lens rather

than as a predefined coding scheme. For example, the emphasis on bridging theory and practice reflects what the authors describe as the epistemological infrastructure of maker-centered learning environments, which draws on constructionist perspectives in which learners actively construct knowledge through engagement with meaningful tasks and artifacts. Furthermore, the importance of group work and collaboration resonates with the social infrastructure of makerspaces, which emphasizes peer learning and shared problem solving. The role of the makerspace itself as a learning environment corresponds to the material–technical infrastructure that enables hands-on experimentation and prototyping. Finally, the project-based course structures and iterative design processes observed in the cases reflect scaffolding structures that support open-ended exploration, experimentation, and learning through iterative design.

5 Conclusion

This case study has examined how maker-centered learning is manifested in the learning outcomes of students across three disciplines when the university makerspace serves as a learning arena. Through our cross-case analysis, we have demonstrated that the makerspace not only provides opportunities for students to acquire technical skills but also fosters broader competences such as collaboration, problem-solving, and reflective understanding. In other words, while the immediate learning outcomes often relate to specific skills such as mastering tools and software, these skills become meaningful within the wider framework of competence development when embedded in group-based, problem-oriented, and iterative design processes.

Our findings indicate that students highlight the bridging of theory and practice as one of the most valuable aspects of their makerspace projects. Iterative design processes, working with materials, and bringing abstract concepts into tangible form all supported a deeper understanding of disciplinary knowledge. Moreover, group work and collaboration, though sometimes experienced as challenging, were reported as important drivers of learning, motivation, and the development of competences that extend beyond technical skills. Importantly, we argue that the creative practice inherent in maker-centered learning is not limited to design- or arts-oriented subjects but also enhances learning in engineering and technology education. Our cases illustrate that creative engagement with materials, tools, and artifacts enhances competence development across disciplinary boundaries. Taken together, the findings suggest that elements of maker-centered learning are evident across the three cases examined in this study.

Future research should further clarify and operationalize the concept of maker-centered learning in higher education. This could involve working with course coordinators to articulate how the approach is embedded in course design and incorporating the concept more explicitly into data collection instruments. A longitudinal design, with data gathered systematically each semester and from a broader sample of students, would also allow for a more nuanced understanding of how learning experiences evolve over time. Such research may help identify not only the benefits of using makerspaces as pedagogical arenas, but also the conditions under which maker-centered learning is most effectively supported.

Data availability statement

The raw data supporting the conclusions of this article will be made available by the authors, without undue reservation.

Ethics statement

The studies involving humans were approved by Sikt—Norwegian Agency for Shared Services in Education and Research. The studies were conducted in accordance with the local legislation and institutional requirements. The participants provided their written informed consent to participate in this study. Written informed consent was obtained from the individual(s) for the publication of any potentially identifiable images or data included in this article.

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